

An otherwise healthy 45 year old man has a vocal cord nodule that the surgeon intends to excise using a CO2 laser. He is a smoker but has no other significant medical history
A. Justify your anaesthetic technique of choice (50%).

My anaesthesia technique of choice is:

- GA with TIVA Propofol.
- Intubation with microlaryngeal tube (MLT) wrapped in laser-resistant material.
- Positive pressure ventilation with NMBA to maintain paralysis.

Other possible techniques include:

1. THRIVE protocol using high flow nasal cannulation (HFNC)
 2. High frequency jet ventilation (HFJV) via rigid bronchoscope.
- These are tubeless techniques which have the advantages of improving surgical access.
 - Patient can be kept on spontaneous ventilation under MAC.
 - Airway topicalisation is necessary to facilitate airway instrumentation.

However, such tubeless techniques have downsides:

Increased risks of desaturation

- FiO₂ is decreased during laser component of surgery
- Sedation is increased for patient's comfort

Loss of airway

- Bleeding
- Tissue edema
- Laryngospasm (higher risk in a smoker)
- Secretions (higher risk in a smoker)

Airway trauma from laser/ surgical manipulation

- Which may be triggered inadvertent coughing/ movement by patient

Hence, my preference would be for GA with secured airway.

GA with TIVA Propofol.

- Allows for titration of depth of anaesthesia independent of gas flows (which will need to be adjusted during laser/ if patient desaturates)
- Less environmental/ air pollution compared to inhalational agents, especially if tubeless technique is used.

Intubation with MLT wrapped in laser-resistant material.

- Secured airway.
- Use of MLT improves surgical access to vocal cord nodule compared to normal sized laser ETT, but not as good as tubeless techniques.
- MLT has single cuff (vs double cuffs in laser ETT) which can be injected with blue dye
- Wrapping MLT in laser resistant material will reduce risk of airway fire, but not completely eliminate it.
- ETT patency is less affected by secretions/ bleeding.
- More accurate monitoring of gas flows and ETCO₂ compared to tubeless techniques.

Positive pressure ventilation with NMBA to maintain paralysis.

- Better control of ventilation.
- Avoidance of inadvertent coughing, which increases risks of airway trauma compared to a patient who is not paralysed
- More responsive adjustment of FiO₂ during laser phase, with ability to rapidly increase FiO₂ and gas flows if patient desaturates.

While I recognise the potential disadvantages of my chosen anaesthesia technique, which include:

- Risks of trauma to vocal nodule during intubation.
- ETT is a potential fuel source in airway fire.
- Poorer surgical access compared to tubeless techniques.

This technique is my preferred choice because it allows for a secured (shared) airway, better control of ventilation/ oxygenation/ gas flows, and still provide surgeon with an immobile operating field for CO₂ laser of vocal cord nodule in a safe manner.